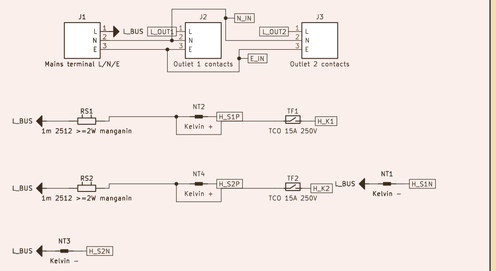
HOT SIDE

Mains referenced. Hot-side ground = L_BUS = LIVE.

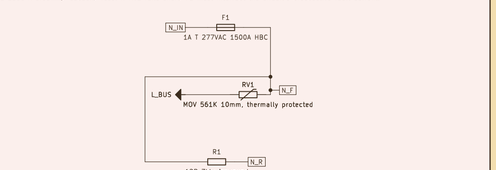
1. MAINS ENTRY AND OUTLET FEEDS

Ring L/N/E land on the busbar; only outlet current passes the shunts and relays.
Shunt common end = L_BUS = hot-side ground. Net ties NT1–NT4 force the IP/IN taps to start at the shunt pads (Kelvin).
Thermal cut-offs: non-resettable; clamped to the outlet feed near the contacts; TF chosen after the EVT thermal test.



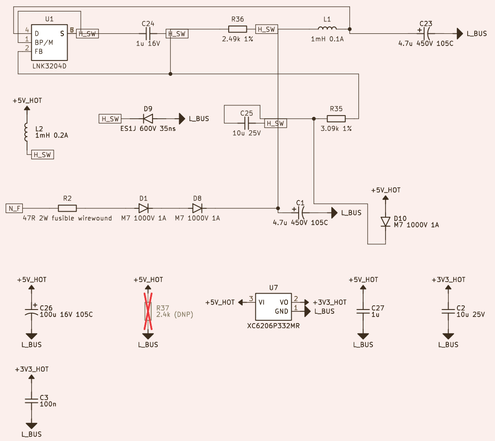
2. ELECTRONICS FEED AND SURGE PROTECTION

Fuse and series resistor per Mornsun's LD03 application circuit. MOV after the fuse, thermally protected type.
F1 is a 1500 A breaking-capacity fuse: if RV1 fails short, F1 must interrupt the circuit's prospective fault current.



3. HOT-SIDE SUPPLY (LNK3204 HALF-WAVE BUCK, 5 V, THEN 3.3 V)

Replaces the KP3310 (R1 study, hardware/sim/R1-study.md): half-wave charging through a surge resistor could not supply the 8–15 mA load.
High-side buck referenced to L_BUS. RF 47R + 2 x M7 + pi filter (2 x 4.7 uF 450 V, 1 mH): simulated bulk/drain peak 465 V at a 4 kV surge (switch rated 725 V).
Feedback: C25 samples Vout during freewheel: 2.00 V x (1 + 3.09k/2.49k) = 4.48 V + about 0.5 V diode offset = 5.0 V (trim R35 at bring-up).
BP/M = 1 uF selects the reduced current limit (enough for 20 mA, allows a small choke). D9 must be <= 35 ns trr (enclosure above 70 C).

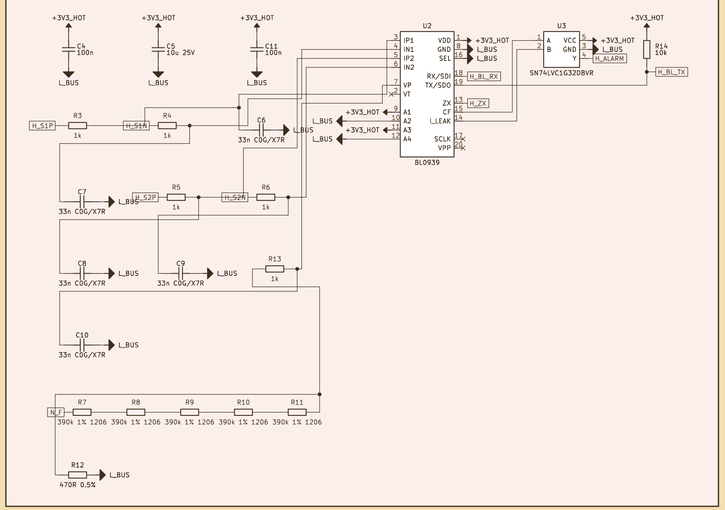


HOT SIDE, SENSING

Metering and wiring checks, all mains referenced.

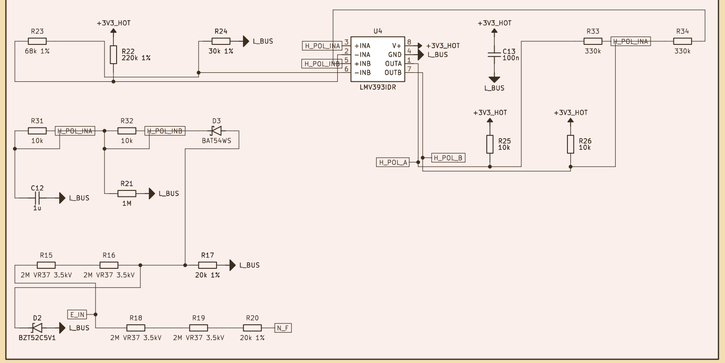
4. METERING (BL0939)

UART address 5 (A1, A3 high; A2, A4 low): the firmware sends 0x55/0xA5, which encode address 5.
Divider 5 x 390k + 470R: 230 V gives 55 mV rms, 290 V gives 70 mV rms (pin limit 70 mV rms). Firmware board file: div_r_bottom = 0.47 k.
Alarms: CF (channel A, MODE[12]) OR !_LEAK (channel B) -> isolator channel E. Both active high.



5. POLARITY AND EARTH CHECK

Arm A: E –> 2 x 2M (HV) –> sense 20k –> L_BUS. Arm B: E –> 2 x 2M (HV) + 20k –> N_F, so a missing earth floats to mid-supply.
Peak detector BAT54WS + 1u // 1M (tau 1 s); thresholds 1.02 V (bit A) and 0.31 V (bit B) from a 220k/68k/30k ladder,
with about 100 mV of hysteresis from R33/R34 so the bits cannot chatter near a threshold.
Correct: A=1 B=1. Swapped L/N: A=0 B=0. No earth: A=0 B=1. Each arm must withstand a 6 kV impulse: VR37-class HV resistors.

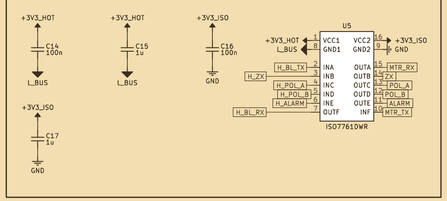


BARRIER CROSSING

The only parts that bridge the barrier.

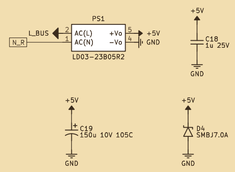
6. DIGITAL ISOLATOR (ISO7761DWR), STRADDLES THE 8 MM BARRIER

Side 1 (hot): BL0939 TX, zero-cross, polarity A/B, alarm in: BL0939 RX out. Side 2 (safe): firmware nets.
Routed slot under the package. Outputs default high if the other side is unpowered: firmware reads POL_x only while the meter is alive.



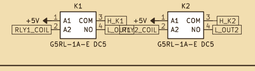
7. SAFE-SIDE 5 V SUPPLY (MORNSUN LD03-23B05R2), STRADDLES THE BARRIER

Genuine Mornsun only. 600 mA, derated to 480 mA at 85 C. Output TVS protects the safe side if the module fails.



8. OUTLET RELAYS, COIL ON THE SAFE SIDE, CONTACTS ON THE HOT SIDE

Live-only switching (ADR-0002). 8 mm between coil and contact pins with a routed slot. Coil 5 V, 80 mA pull-in.



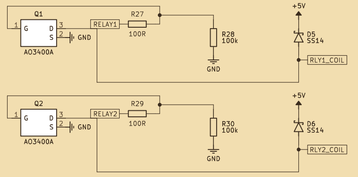
REINFORCED BARRIER, 8 mm

SAFE ZONE

SELV.

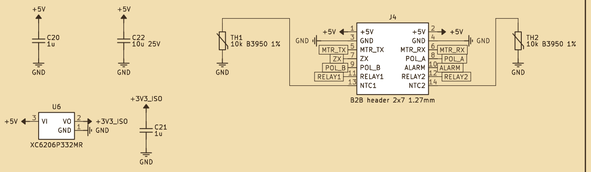
9. RELAY DRIVERS

LEDC PWM at 20 kHz: 100% for 50 ms, then the holding duty. 100k keeps the coil off while the ESP32 boots.



10. ISOLATOR 3.3 V, RELAY NTCS, BOARD-TO-BOARD HEADER

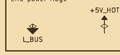
XC6206 feeds only the isolator's safe side (keeps 3.3 V logic levels to the ESP32). NTCs are pressed against each relay body;
their 10k pull-ups sit on the front board next to the ADC.



BOARD NOTES

REAR BOARD, EVT1 DRAFT FOR REVIEW
2 layers, 2 oz. One reinforced barrier: 8.0 mm creepage and clearance, crossed only by U5, PS1, K1, K2.
Net classes: MAINS = L_BUS, N_IN, E_IN, N_F, N_R, L_OUT1/2 and every H_* net. SAFE = everything else.
MAINS to SAFE >= 8.0 mm; MAINS to E_IN >= 3.0 mm; L to N >= 3.0 mm with slots where traces converge.
Outlet paths sized for 20 A combined (IPC-2152). No connector on the hot side.
Bench rule: never power this board from the mains without an isolation transformer and an RCD.

ERC power flags



Hot side is referenced to LIVE (L_BUS). Treat every hot-side net as mains.
Smart Wall Socket. DRAFT for engineering review, not a release.

Aether Labs

Sheet: /
File: sws-rear.kicad_sch

Title: Smart Wall Socket EVT1, rear board

Size: User	Date: 2026-11-10	Rev: EVT1-draft-D
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1